Lifetime Sequence-of-Returns Risk

As we are starting to see, individual investors are vulnerable to the sequence of market returns experienced over their investing lifetimes. Individuals who behave in exactly the same way over their careers—saving the same percentage of the same salary for the same number of years—can experience disparate outcomes based solely on the specific sequence of investment returns that accompanies their career and retirement. Peak vulnerability is reached at the retirement date, when returning to the labor force becomes increasingly difficult and a postretirement market drop can be devastating. Actual wealth accumulations and sustainable withdrawal rates will vary substantially among retirees, as these outcomes depend disproportionately on the shorter sequence of returns just before and after the retirement date.

In other words, individuals are the most vulnerable when their wealth is likely the largest it has ever been, in absolute terms, due to the sequence of returns. Two investors may enjoy the same average return on their investments but could still experience divergent outcomes based on the sequence in which returns arrive. This can impact both those who are saving and contributing to their portfolio, and those who are withdrawing a constant stream of cash flows from their portfolio.

Historical simulations based on overlapping periods reveal how sequence of returns can create differing outcomes for otherwise identical investors. We'll look at the life of one individual investor we'll call Adam. The only unknown Adam faces with regard to his retirement planning is what his specific sequence of market returns will be. This simplifies reality, as Adam is not saddled with uncertainty in his future employment status and salary.

Adam saves for retirement during the final thirty years of work, and he earns a constant real income in each of these years. For thirty years, he puts away a fixed savings rate of 15 percent of his income at the end of each year. The wealth accumulation achieved at retirement is defined as a multiple of Adam's constant real salary. In other words, if the wealth accumulation is 10×, then Adam had savings equal to ten times his salary upon reaching retirement. A $100,000 salary means a $1 million portfolio. In this simplified